Unit Two

Disorders of the Respiratory system

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Anatomy and physiology overview

- The respiratory system is composed of the upper and lower respiratory tracts.
- **Upper airway** structures consist of the nose, sinuses and nasal passages, pharynx, tonsils and adenoids, larynx, and trachea.
- **Lower respiratory tract** consists of the lungs, which contain the bronchial and alveolar structures needed for gas exchange.

Upper respiratory tract

- ✓ **Nose:** is composed of an external and an internal portion.

- The anterior nares (nostrils) are the external openings of the nasal cavities.

- The internal portion of the nose is a hollow cavity separated into the right and left nasal cavities by a narrow vertical divider, the septum.

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- The nasal cavities are lined with highly vascular ciliated mucous membranes called the nasal mucosa. It contains fine hairs inside, the cilia.
- The nose serves as a passageway for air to pass to and from the lungs.
- It filters impurities and humidifies and warms the air as it is inhaled.

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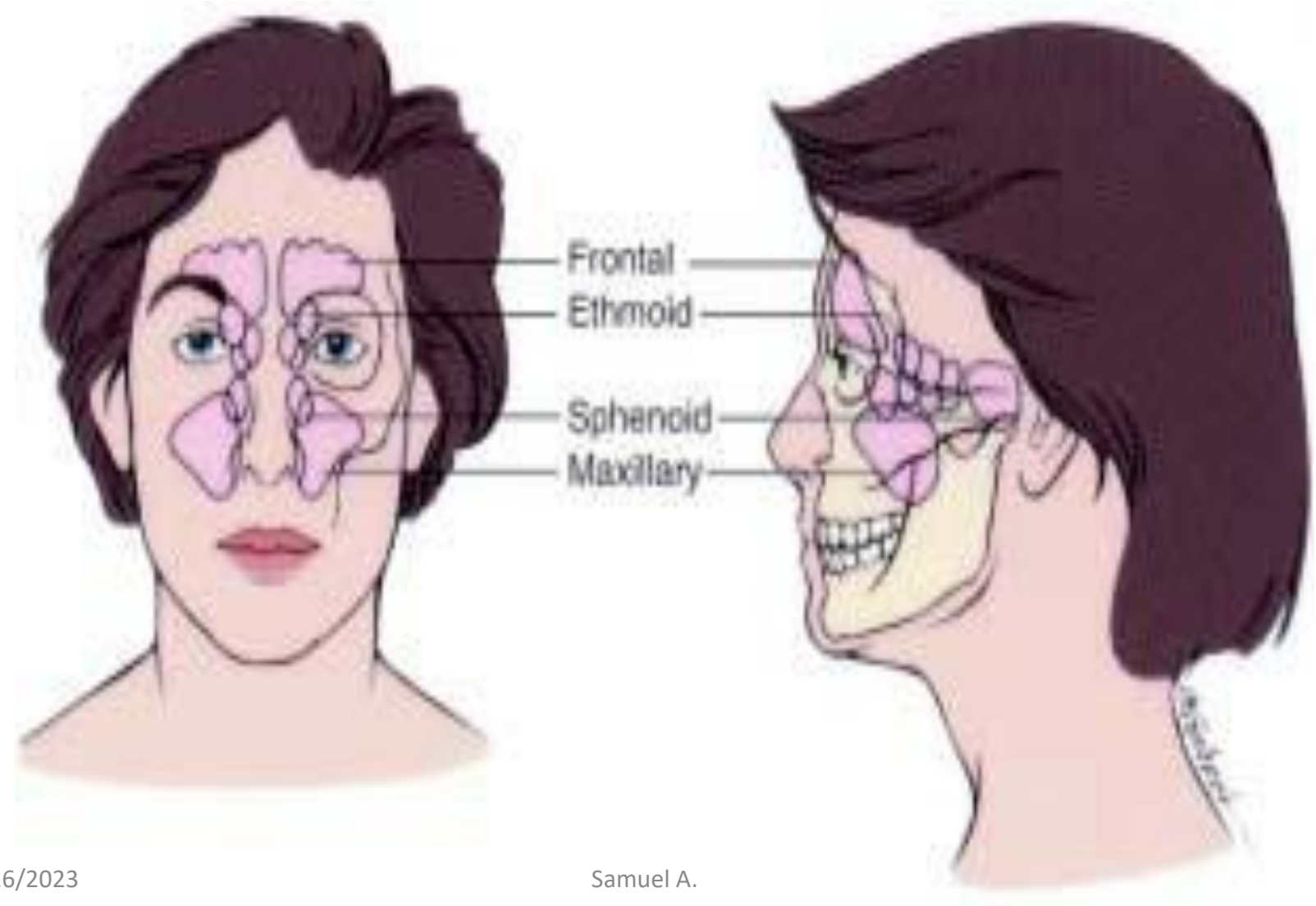
✓ Para nasal Sinuses

- The Para nasal sinuses include four pairs of bony cavities

- The sinuses are named by their location: **frontal, ethmoidal, sphenoid,** and **maxillary**

- A prominent function of the sinuses is to serve as a resonating chamber in speech.

Fig. The sinuses



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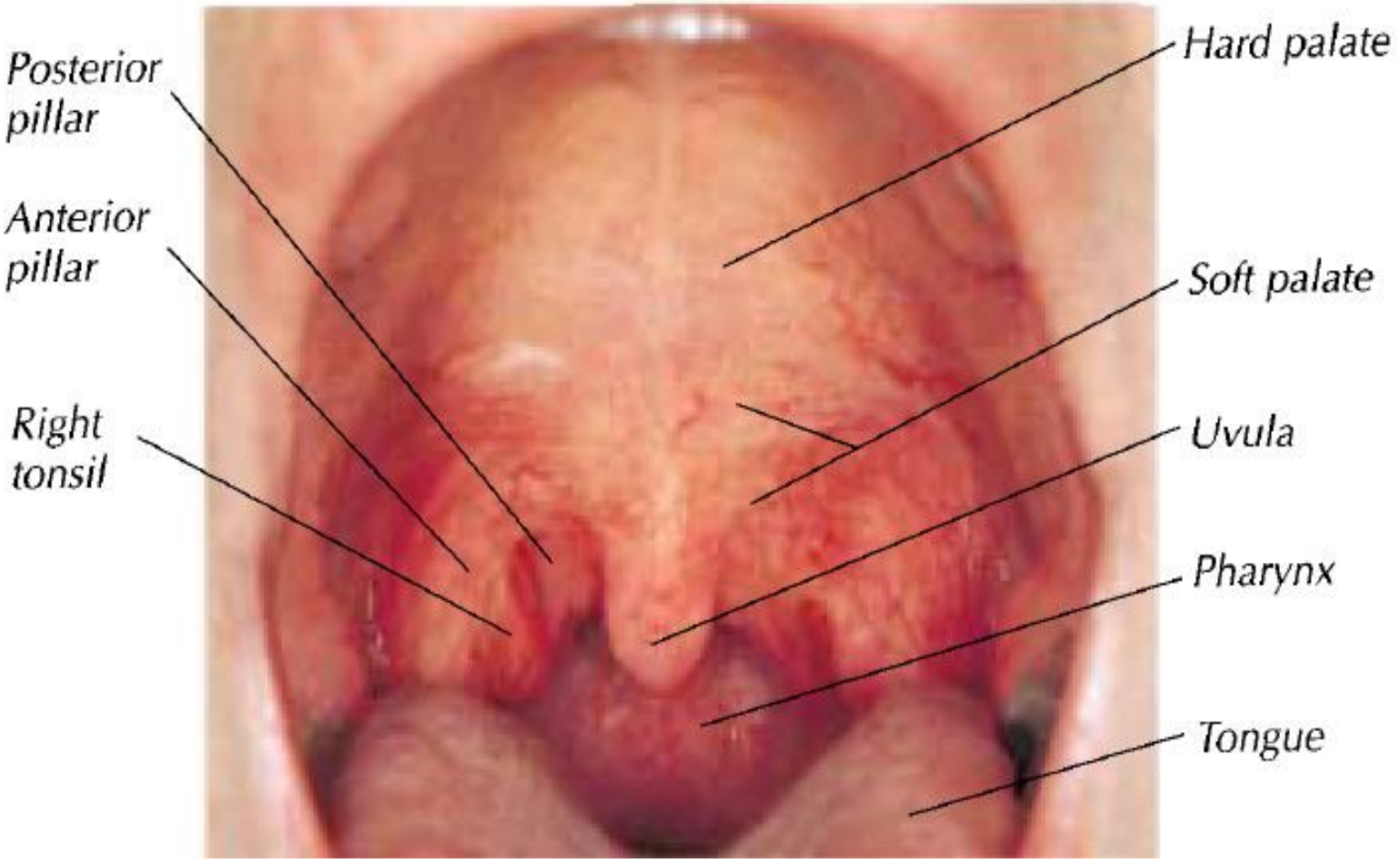
✓ Pharynx, Tonsils, and Adenoids

- The pharynx, or throat, is a tube like structure that connects the nasal and oral cavities to the larynx.
- The adenoids, or pharyngeal tonsils, are located in the roof of the nasopharynx.
- The tonsils, the adenoids, and other lymphoid tissue encircle the throat.

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- These structures are important links in the chain of lymph nodes guarding the body from invasion by organisms entering the nose and the throat
- The pharynx functions as a passage way for the respiratory and digestive tracts.

Fig. The pharynx and other oral structures



Cont---

✓ Larynx

- The larynx, or voice organ, is a cartilaginous epithelium-lined structure that connects the pharynx and the trachea.

- The major function of the larynx is vocalization. It also protects the lower airway from foreign substances and facilitates coughing.

Cont---

✓ Trachea

- The trachea, or windpipe, is composed of smooth muscle with C-shaped rings of cartilage at regular intervals.
- It serves as the passage between the larynx and the bronchi.
- In general the primary function of upper respiratory tracts are to warm, humidify, and filter entering air and to protect the lower airway from foreign material.

Lower respiratory tract

✓ Lungs

- The lungs are located within the thoracic cavity and it is surrounded by ribs & muscles.
- The lungs and wall of the thorax are lined with a serous membrane called the pleura.
- The lungs are covered by the two thin membranes: i.e. the visceral pleura covers the lungs; the parietal pleura lines the thorax.

Cont---

- The space b/n visceral and parietal pleura called pleural cavity/space.
- The space contains small amount of serous fluid .
- Each lung is divided in to lobes i.e. the left lung consists of upper and lower lobes and the right lung upper, middle and lower lobes.

Cont---

✓ Bronchi and Bronchioles

- There are several divisions of the bronchi within each lobe of the lung.
- First are the lobar bronchi (three in the right lung and two in the left lung).

Lobar bronchi divide into segmental bronchi (10 on the right and 8 on the left

- Segmental bronchi then divide into sub segmental bronchi.

Cont----

- The sub segmental bronchi then branch into bronchioles,

✓ **Alveoli**

- The lung is made up of about 300 million alveoli, which are arranged in clusters of 15 to 20.

Fig. Respiratory tracts

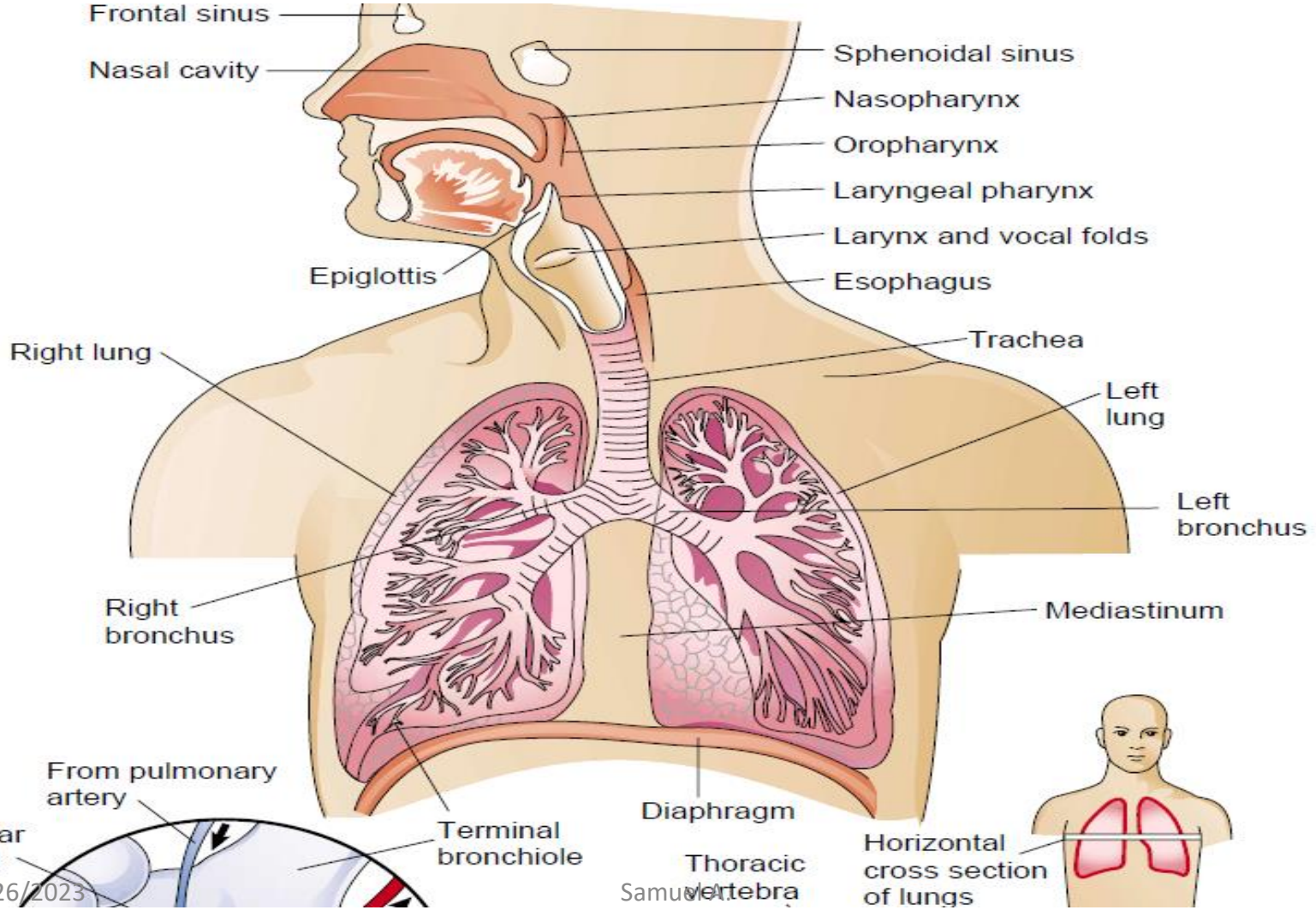
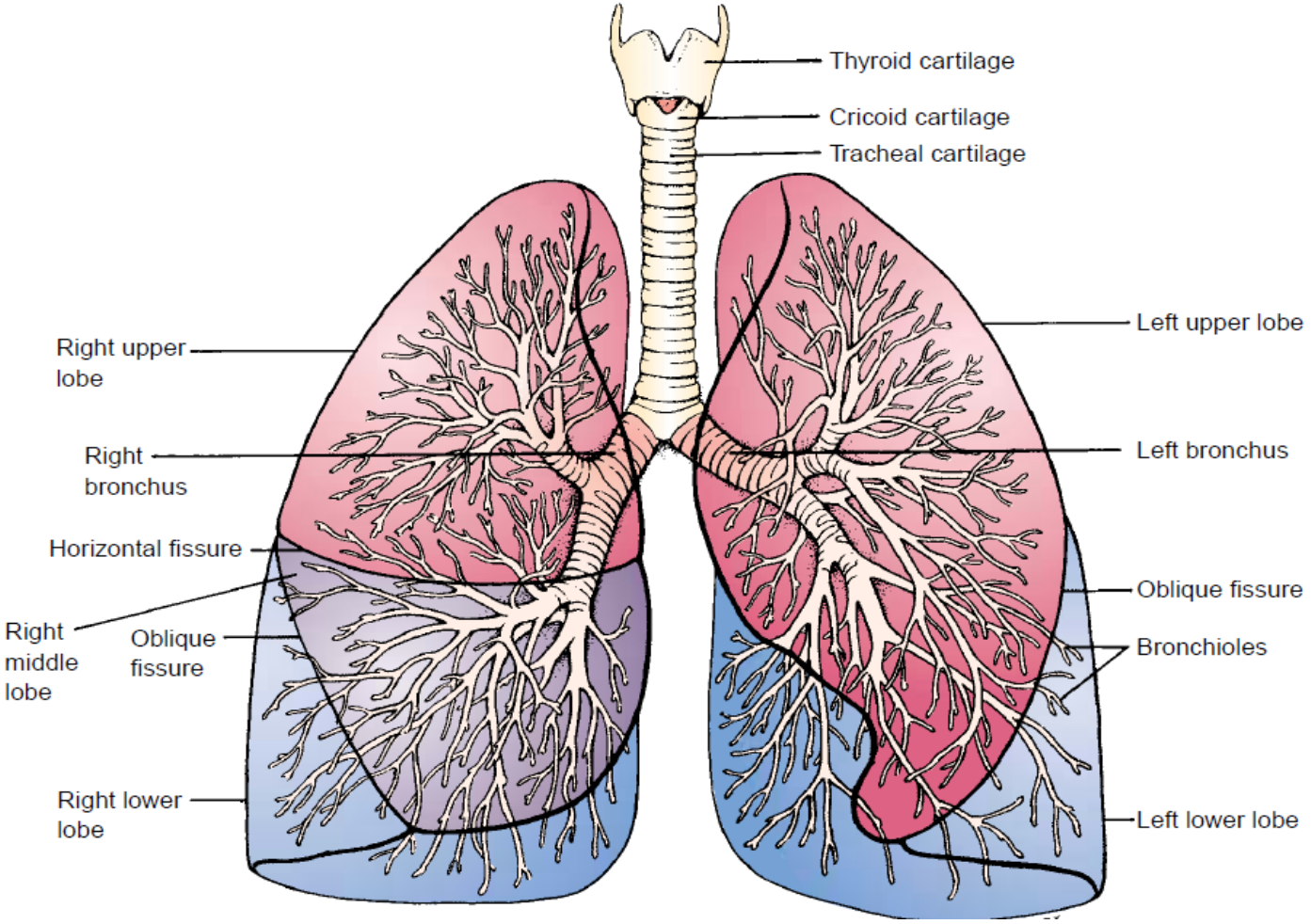


Fig. The lobes of the lung



Function of respiratory system

- The respiratory system performs life-sustaining processes such as:
  - Oxygen transport,
  - Respiration (process of gas exchange between the atmospheric air and the blood, and between the blood and cells of the body is called **respiration**.)
  - Ventilation (air flow in and out of lungs),
  - Gas exchange.

Lung Volumes and Capacities

- Lung volumes are categorized as tidal volume, inspiratory reserve volume, expiratory reserve volume, and residual volume.

- Lung capacity is evaluated in terms of vital capacity, inspiratory capacity, functional residual capacity, and total lung capacity.

- **Tidal volume (TV):** The volume of air inhaled and exhaled with each breath, amount (500ml)

- **Inspiratory reserve volume (IRV)** : The maximum volume of air that can be inhaled after a normal inhalation (3000ml)

- **Expiratory reserve volume (ERV):** The maximum volume of air that can be exhaled forcibly after a normal exhalation (1,100ml)
- **Residual volume(RV) :** The volume of air remaining in the lungs after a maximum exhalation (1,200ml)

- **Inspiratory capacity (IC)** The maximum volume of air inhaled after normal expiration (3500ml)

  IC = TV + IRV

- **Vital capacity (VC):** The maximum volume of air exhaled from the point of maximum inspiration (4,600ml)

  VC = TV + IRV + ERV

- **Functional residual capacity (FRV):** The volume of air remaining in the lungs after a normal expiration (2300ml)

  FRV = ERV + RV

- **Total lung capacity (TLC):** The volume of air in the lungs after a maximum inspiration (5,800ml)

  TLC = TV + IRV + ERV + RV

Assessment: Health history

- The health history focuses on:
  - the physical and functional problems of the patient
  - the effect of these problems on his or her life.
- The reason the patient is seeking health care often is related to one of the following:
  - Dyspnea (shortness of breath)
  - pain
  - accumulation of mucus
  - wheezing
  - hemoptysis (blood spit up from the respiratory tract)
  - edema of the ankles and feet
  - cough
  - general fatigue and weakness.

Cont----

- In addition to identifying the chief reason why the patient is seeking health care, the nurse collects information about:-
  - *precipitating factors*
  - *duration*
  - *severity*
  - *associated factors* or symptoms and
  - also assesses for *risk factors* and genetic factors

Cont----

- **Risk Factors for Respiratory Disease include:**
  - Smoking (the single most important contributor to lung disease)
  - Personal or family history of lung disease
  - Occupation
  - Allergens and environmental pollutants
  - Recreational exposure

- **Common Signs and Symptoms**

  The major signs and symptoms of respiratory disease are:
  - *dyspnea (difficult or labored breathing)*
  - *cough*
  - *sputum production*
  - *chest pain*
  - *wheezing*
  - *clubbing of the fingers*
  - *hemoptysis*, and
  - *cyanosis*.

- These clinical manifestations are related to the duration and severity of the disease.

Assessment: Physical Examination

PHYSICAL ASSESSMENT OF THE UPPER RESPIRATORY STRUCTURES

- Inspect and palpate nose and sinuses
- Assess the pharynx and mouth
- Note the position of trachea by direct palpation

- **PHYSICAL ASSESSMENT OF THE LOWER RESPIRATORY STRUCTURES**

Inspection of Thorax

- Inspection of the thorax provides information about the musculoskeletal structure, the patient's nutritional status, and the respiratory system.

- Observe the presence or absence of the following:

- *Chest configuration*: normally, the ratio of antero posterior diameter to lateral diameter is 1:2.

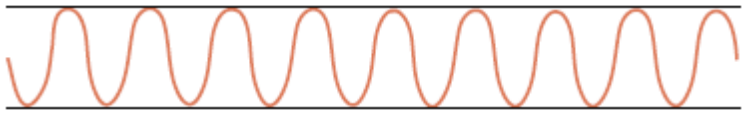
- But it can change due to the following deformities:-

  - ✓*Barrel chest*
  - ✓*Funnel chest (pectus excavatum)*
  - ✓*Pigeon chest (pectus carinatum)*
  - ✓*Kyphscoliosis*

BREATHING PATTERNS AND RESPIRATORY RATES

- Observing the rate and depth of respiration is a simple but important aspect of assessment.

- **Eupnea:** Normal, breathing at 12- 20 breaths/minute

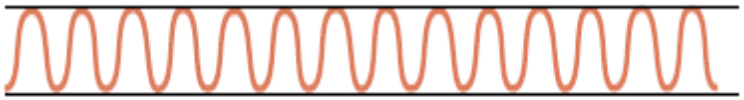


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- ✓ **Bradypnea:** Slower than normal rate (< 10 breaths/minute), with normal depth and regular rhythm.

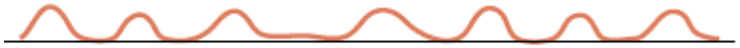


- ✓ **Tachypnea:** Rapid, shallow breathing > 24 breaths/minute

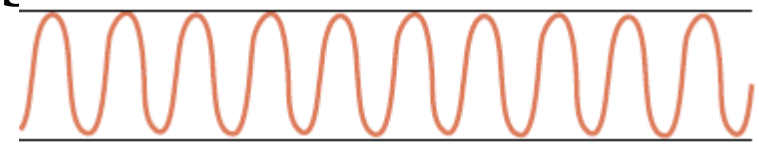


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- ✓ **Hypoventilation:** Shallow, irregular breathing



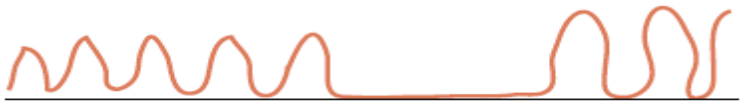
- ✓ **Hyperventilation:** Increased rate and depth of breathing (called Kussmaul's respiration if caused by diabetic ketoacidosis)



- Hyperventilation that is marked by an increase in rate and depth, associated with severe acidosis of diabetic or renal origin, is called ***Kussmaul's respiration***.

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- ✓ **Apnea:** Period of cessation of breathing. Time duration varies; apnea may occur briefly during other breathing disorders, such as with sleep apnea. Life threatening if sustained.

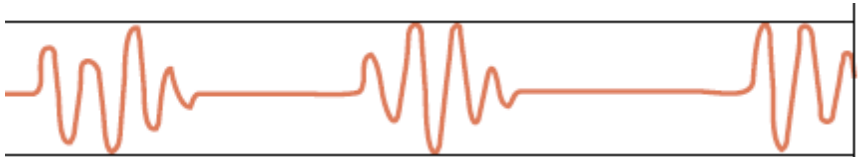


- ✓ **Cheyne-Stokes:** Regular cycle where the rate and depth of breathing increase, then decrease until apnea (usually about 20 seconds) occurs.



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- **Biot's respiration:** Periods of normal breathing (3-4 breaths) followed by a varying period of apnea (usually 10 seconds to 1 minute).



Thoracic Palpation

- The nurse palpates the thorax for:
  - Tenderness,
  - Masses,
  - Lesions,
  - Respiratory excursion (an estimation of thoracic expansion), and
  - Vocal fremitus (vibration produced in the chest by sound generated in the larynx).

Thoracic Percussion

- Percussion sets the chest wall and underlying structures in motion, producing audible and tactile vibrations.

- The nurse uses percussion to determine whether underlying tissues are filled with air, fluid, or solid material.

- Percussion also is used to estimate the size and location of certain structures within the thorax (eg, diaphragm, heart, liver).

Thoracic Auscultation

- Auscultation is useful in assessing the flow of air through the bronchial tree and in evaluating the presence of fluid or solid obstruction in the lung structures.

- The nurse auscultates for normal breath sounds, adventitious sounds, and voice sounds.

Common diagnostic tests for respiratory disorders

- **Pulmonary function tests (PFTs):** measurements of lung volumes, ventilatory function, and the mechanics of breathing, diffusion, and gas exchange. They are routinely used in patients with chronic respiratory disorders.

- **Arterial blood gas studies:** Measurements of blood pH and of arterial oxygen tension (PaO2) and carbon dioxide tension (PaCO2) are obtained when managing patients with respiratory problems and in adjusting oxygen therapy as needed.

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- **Pulse Oximetry:** a noninvasive method of continuously monitoring the oxygen saturation of hemoglobin. Normal values are 95% to 100%.

- Values less than 85% indicate that the tissues are not receiving enough oxygen, and the patient needs further evaluation

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- **Cultures:** Throat cultures may be performed to identify organisms responsible for pharyngitis.

- It may also assist in identifying organisms responsible for infection of the lower respiratory tract.

- Nasal swabs also may be performed for the same purpose.

Cont---

- **Sputum studies:** Sputum is obtained for analysis to identify pathogenic organisms and to determine whether malignant cells are present.

- Sputum cultures are used in diagnosis, for drug sensitivity testing, and to guide treatment.

Imaging studies

✓ **Chest X-Ray:** Normal pulmonary tissue is radiolucent; therefore, densities produced by fluid, tumors, foreign bodies, and other pathologic conditions can be detected by x-ray examination.

- A chest x-ray may reveal an extensive pathologic process in the lungs in the absence of symptoms.

- The routine chest x-ray consists of two views the **posteroanterior** projection and the **lateral** projection.

✓ **Computed Tomography:** an imaging method in which the lungs are scanned in successive layers by a narrow-beam x-ray.

- The images produced provide a cross-sectional view of the chest.

- CT may be used to define pulmonary nodules and small tumors adjacent to pleural surfaces that are not visible on routine chest x-ray, and to demonstrate mediastinal abnormalities and hilar adenopathy, which are difficult to visualize with other techniques.

- ✓ **Magnetic Resonance Imaging:** are similar to CT scans except that magnetic fields and radiofrequency signals are used instead of a narrow-beam x-ray.

- It yields a much more detailed diagnostic image than CT scans.

- MRI is used to characterize pulmonary nodules, stage bronchogenic carcinoma (assessment of chest wall invasion), and evaluate inflammatory activity in interstitial lung disease, acute pulmonary embolism, and chronic thrombolytic pulmonary hypertension

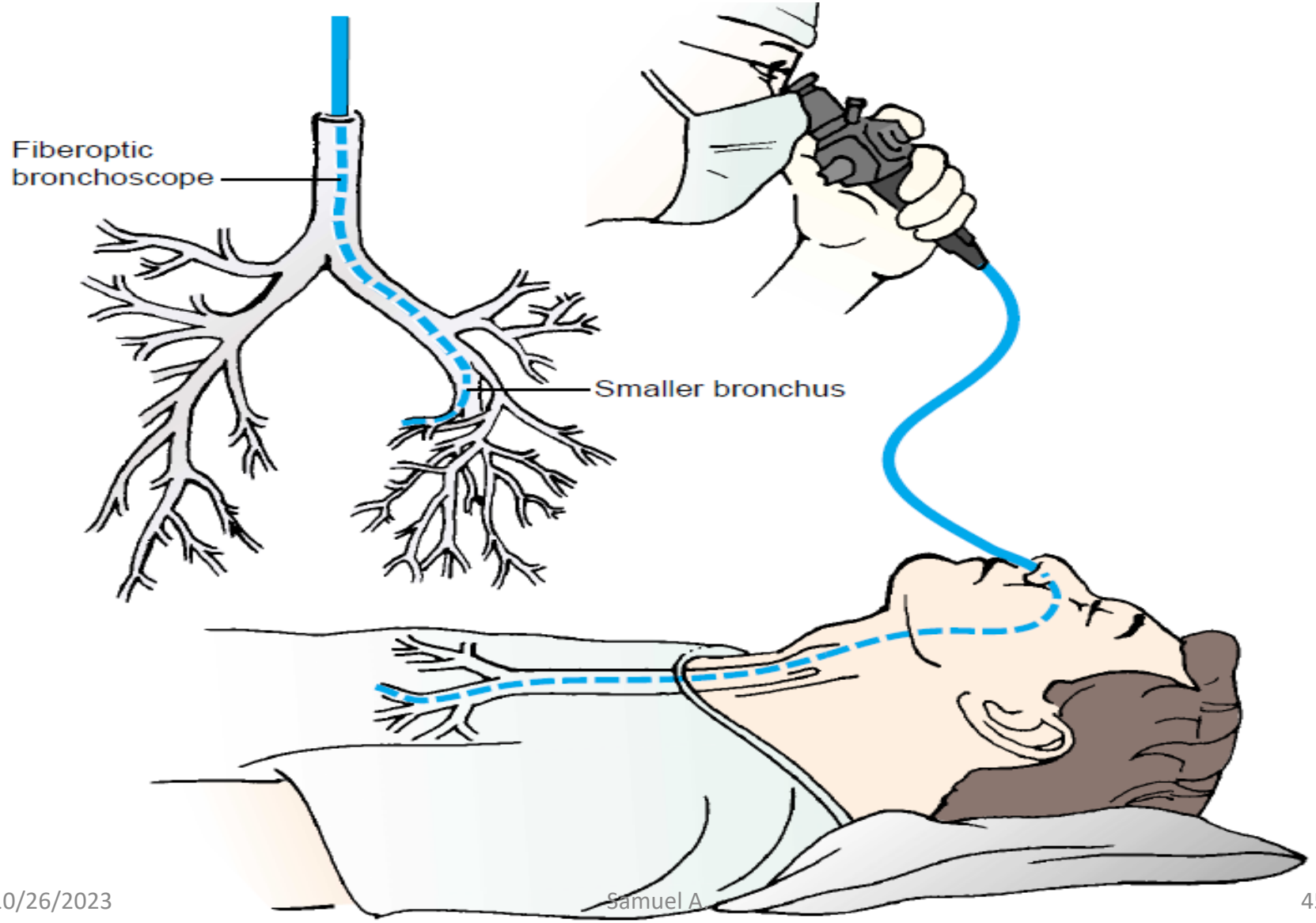
Endoscopic procedures

✓ **Bronchoscopy:** is the direct inspection and examination of the larynx, trachea, and bronchi through either a flexible fiberoptic bronchoscope or a rigid bronchoscope.

- The fiberoptic scope is used more frequently in current practice.

- Bronchoscopy permits the clinician not only to diagnose but also to treat various lung problems.

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- **Thoracoscopy:** is a diagnostic procedure in which the pleural cavity is examined with an endoscope. Small incisions are made into the pleural cavity in an intercostal space; the location of the incision depends on the clinical and diagnostic findings.

- It is primarily indicated in the diagnostic evaluation of pleural effusions, pleural disease, and tumor staging.

- A combined diagnostic–treatment procedure, thoracoscopy includes excising tissue for biopsy.

- Biopsies of the lesions can be performed under visualization for diagnosis.

Thoracentesis

- A thin layer of pleural fluid normally remains in the pleural space.
- An accumulation of pleural fluid may occur with some disorders.
- A sample of this fluid can be obtained by thoracentesis (aspiration of pleural fluid for diagnostic or therapeutic purposes).
- Studies of pleural fluid include Gram's stain, culture and sensitivity, acid-fast staining and culture, differential cell count, cytology, pH, specific gravity, total protein, and lactic dehydrogenase.

- **Biopsy:** the excision of a small amount of tissue, may be performed to permit examination of cells from the pharynx, larynx, nasal passages lymph node and pleural/lungs

- Local, topical, or general anesthesia may be administered, depending on the site and the procedure

UPPER RESPIRATORY TRACT DISORDERS

Upper Airway Infections

1. The Common Cold (Acute coryza)

- It is an acute, usually afebrile, viral infection of the respiratory tract, with inflammation in any or all airways, including the nose, paranasal sinuses, throat, larynx, and often trachea and bronchi.

Etiology

- Many viruses cause the common cold including Picornavirus (rhinoVirus), Influenza and para influenza viruses, Respiratory syncytial virus, Corona- and adeno virus group.

- Infections may be facilitated by excessive fatigue, emotional distress, or allergic naso pharyngeal disorders and during the mid-phase of the menstrual cycle.

Signs and Symptoms

- Onset is abrupt after a short (1 to 3 days) incubation period.
- Illness generally begins with nasal or throat discomfort followed by sneezing, rhinorrhea, and malaise.
- The disease is afebrile and pharyngitis is usually present.
- Nasal secretions, watery and profuse during 1st or 2nd day of symptoms, become more mucous and purulent.
- Hacking cough associated with scanty sputum often lasts into the 2nd week.
- When no complications occur, symptoms normally resolve in 4 to 10 days.

Diagnosis

- Symptoms and signs are nonspecific
- Bacterial infections, allergic rhinorrhea, and other disorders also cause upper respiratory tract symptoms at onset.
- Differentiation depends on the season and the course of the Symptoms
- Fever more severe symptoms usual indicate influenza.

Treatment

- A Warm, comfortable environment and measures to prevent direct spread of infection are recommended for all persons.
- Antipyretics and analgesics are given to control fever and malign.
- Nasal decongestants are used if patients have nasal congestion.
- Steam inhalation is also used in nasal congestions to help mobilize secretions and relieve chest tightness.

Treatment ...

- Treat persistent cough with cough suppressants.

- Antibiotics are not effective against viruses, so they are not recommended unless a specific bacterial complication develops

2. Tonsillitis

- It is an inflammation & enlargement of the tonsil tissue
- It can be acute or chronic

- Tonsil frequently serve as the site of acute infection b/c tonsils filter organisms and protect the respiratory tract from infection.

2.1. Acute tonsillitis

Causative agent

- Bacteria: /the most common
- Others : staphylococcus, H. influenza and others
- It can be viral in origin
- It is very common up to the age of 15years.
- It can occur at any age but not common in adults.

Predisposing factor

- URTI
- Lowered immunity
- Pollution
- Hot or cold temperatures

Clinical Features

- Sore throat
- Dysphagia
- Headaches
- Fever
- Red and pus or exudates present on the tonsils.

Diagnostic evaluation

- History
- Physical examination
- Culture & sensitivity.

Medical Mgt

- If bacteria:

✓ First line

  - Amoxicillin 500mg po. tid for 7 days OR
  - Ampicillin 500mg QID FOR 7days or
  - Benz. Penicillin 2.4 Million Iu. Im. stat or procaine penicillin 800, 000 IU IM daily for 5-7 days

✓ Alternative

- Erytomycin 250 mg every 6 hrs for 5 days

- If viral only asymptomatic therapy

Nursing intervention

- Analgesics such as paracetamol 500mg Po PRN
- Tepid sponge
- Bed rest and soft diet
- Warm saline gargles

Complication

- Laryngeal edema
- Sinusitis (inflammation of the mucus membrane of one or more sinuses)
- Acute rhinitis (inflammation of the nasal mucosa)
- Acute otitis media
- Septicemia
- Chronic tonsillitis

2.2 Chronic tonsillitis

- It is due to recurrent or unresolved acute tonsillitis.

Etiology

- Similar to acute tonsillitis

Clinical manifestation

- Mild edema, scaring of the tonsil and pretonsilar abscess.
- Scratching sensation in the throat , cough,
- Recurrent pain
- Cough
- Enlarged cervical lymph node

Treatment

- Surgery/ tonsillectomy

- Nutritious diet (high liquid and semi liquid diet for several days, avoid spicy, cold, acidic or rough foods) & vitamins

- Antibiotic & Analgesics

3. Pharyngitis

- Inflammation of the Pharynx secondary to an infectious agent
- Most common infectious agents are Group A Streptococcus and various viral agents
- It can also caused by inhalation of irritant gases or ingestion of irritant liquid.
- Often co-exists with tonsillitis

3.1. Acute Pharyngitis

Causes

- Most causes are viruses

- Bacterial:
  - Group A beta-hemolytic streptococcus (most common bacteria cause)
  - Mycoplasma pneumoniae
  - Arcanobacterium haemolyticum
  - Neisseria gonorrhea
  - Chlamydia pneumoniae

Clinical Features

- Fiery-red pharyngeal membrane & tonsil
- Lymphoid follicles that are swollen and flecked with white-purple exudate
- Enlarged & tender cervical lymph nodes
- Fever, malaise, & sore throat

Diagnosis

- Clinical sign & symptoms plus
- Throat culture

Differential diagnosis

- Tonsillitis
- laryngitis

Complication

- Otitis Media
- Mastoditis
- Sinusitis
- Peritonsilar abscess. Etc

Medical Rx:

- Viral subsides within 10days if not complicated,
- Viral pharingitis does not improve with any chemotherapy

✓ For bacterial causes

- Penicillin-for group A beta-hemolytic streptococci
- Cephalosporin's and macrolides (clarithromycin and azithromycin) may be used.
- Analgesic

Nursing Intervention

- Instruct bed rest during febrile stage of illness.
- Liquid or soft diet is provided during acute stage of the disease.
- Warm saline gargles or irrigation are used. Irrigating the throat properly is an effective means of reducing spasm in the pharyngeal muscle.
- Mouth care can be given.

3.2. Chronic Pharingitis

- Is a persistent inflammation of the pharynx.
- It is common in adults:
  - who work or live in dusty surroundings
  - use their voice to excess
  - suffer from chronic cough, and
  - habitually use alcohol and tobacco.

- Three types of chronic pharyngitis are recognized:

  ✓Hypertrophic

  ✓Atrophic

  ✓Chronic granular

Clinical Manifestations

- A constant sense of irritation or
- Fullness in the throat,
- Mucus that collects in the throat and can be expelled by coughing, and
- Difficulty swallowing.

- Treatment of chronic pharyngitis is based on:
  - relieving symptoms,
  - avoiding exposure to irritants, and
  - correcting any upper respiratory, pulmonary, or cardiac condition that might be responsible for a chronic cough.

4. Laryngitis

- It is an inflammation of the larynx often occurs as a result of voice abuse, exposure to dust, chemicals, smoke, & other pollutant.

Etiology

- Almost always virus

- Bacterial invasion may be secondary.

Risk factors

- URTI
- Vocal misuse and over use
- Irritation / smoking, alcohol /
- Seasonal changes /sudden temperature change
- Iatrogenic / intubations and endo-laryngeal surgery /
- Laryngitis is usually associated with acute rhinitis or naso-pharyngitis.

Clinical Features

- Hoarseness or complete loss of voice /aphonia
- Severe cough
- Pain occurs in severe cases
- Stridor may be present in children
- Edema, exudates, congestion
- Laryngitis can be the complication of chronic sinusitis & chronic bronchitis.

Medical Management

- Resting the voice
- Avoid smoking
- Bed rest
- Inhaling cool steam or aerosol
- Treat secondary bacterial infection with antibiotics.

Nursing process for the patient with upper respiratory infection

✓ Assessment

- A complete history
- Inspection / throat, nose, and tonsils/
- Palpate / mass, deformity, tracheal position, cervical lymph nodes /

✓ Nursing diagnosis may include

- Ineffective airway clearance related to excessive secretion to inflammatory process.
- Pain related to irritation secondary to infection
- Impaired verbal communication related to irritation secondary to infection.
- Fluid volume deficit related to increased fluid loss secondary to diaphoresis associated with fever
- Knowledge deficit regarding prevention, drugs regimen

✓ Planning

- Maintenance of a patient airway
- Relief of pain
- Maintenance of effective means of communication
- Absence of fluid volume deficit
- Knowledge of how to prevent infection

✓ Nursing Intervention

- Clearing airway:
  - Increase fluid intake
  - Humidifying the environment /inhaling steam/.
- Promote comfort Measures : Give analgesics /
- Promoting communication: Voice rest
- Encouraging fluid intake
- Patient teaching – treatment regimen, prevention
- Monitor & Manage potential complication

✓ Evaluation

❖ Expected Outcomes

- Maintains a patient airway by managing secretion
- Report feeling more comfortable
- Maintain adequate fluid
- Becomes free of signs & symptoms

LOWER RESPIRATORY TRACT DISORDERS

Bronchitis/Acute tracheo-bronchitis

- Is an acute inflammation of the mucous membranes of the trachea & the bronchial tree.

Causative agents

- It is often virus but the common bacterial causes are:-
    - Streptococcus pneumaniae
    - Haemephilus Influenza
    - Mycoplasma pneumonia
    - Often follows infections of the upper respiratory tract.

N.B. Inhalation of physical e.g. microorganisms & chemical irritants, gases & other air contaminants can also causes bronchitis

Predisposing factors

- Pre- existing URTI

Clinical Manifestations

- Initially, that patient has a dry, irritating cough & expectorates scanty mucoid sputum

- As the infection progresses, inspiration *and expiration* may became noisy (inspiratory *stridor* and expiratory *wheeze*) & cough with more profuse purulent /thick sputum may be present.

- Others: Fever, headache, & generalized malaise.

Medical Management

- Antibiotic Treatment:-
  - It is indicated, when bronchitis is complicated by bacterial infections.

✓ **First line Antibiotics**

- Ampicillin 500mg P.O Qid, for 7 days OR
- Amoxicillin 500gm P.O tid, for 7 days

✓ **Alternative Antibiotic**

- Erythromycin 500mg P.O Qid for 7 days

  OR
- Tetracycline 500mg Qid, for 5-7 days

  OR
- Co - trimoxazole 480 mg 2 tabs po bid for 7 days

Nursing Interventions

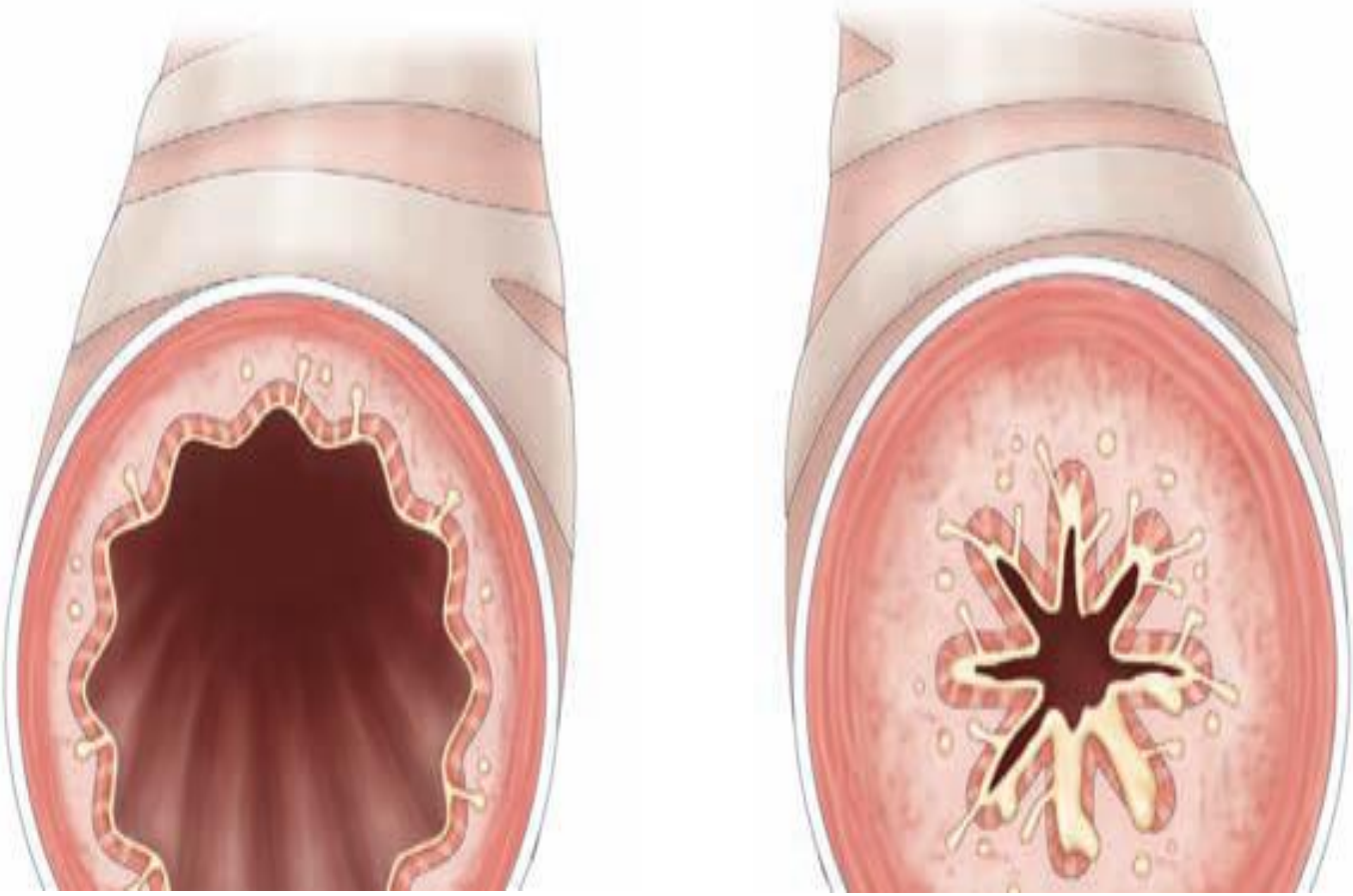
- Encourage frequent coughing to remove secretions
- Advice bed rest
- Steam inhalation
- Advice fluid intake to thin the viscous & tenacious secretions.

✓Chronic bronchitis

- It is the presence of a productive cough that lasts 3 months a year for two consecutive years, in the absence of major lung disease (WHO).

- The accumulated secretions in the bronchioles interfere with effective breathing.

NORMAL BRONCHUS **CHRONIC BRONCHITIS**



Causes

- Cigarette smoking
- Exposure to pollution
- Patients with chronic bronchitis are more susceptible to recurring infections of the lower respiratory tract.

Clinical Manifestations

- A chronic, productive cough in the winter months (recurrent coughing and sputum production)
- History of cigarette smoking & frequent respiratory infection
- Production of thick, gelatinous sputum (greater amount's produced during super imposed infection)
- Wheezing and dyspnea as disease progresses
- Recurrent acute respiratory infections followed by persistent cough

Diagnostic Evaluation

- A complete history, including family, environmental exposure to irritating substance, & occupational history, and also history of smoking (number of packs per day)
- Physical examination
- Chest x-ray
- Lung function studies
- Arterial blood gas analysis

Medical Management

- Remove bronchial secretions
- Bronchodilators are prescribed to relieve bronchospasm & reduce air way obstruction.
- Postural drainage
- Increase fluid intake
- The patient must stop smoking

Bronchiectasis

- Is a chronic & permanent/irreversible dilatation of the bronchi and bronchioles due to destruction of bronchial wall.
- The dilatation is due to destructive & Inflammatory changes in the wall of medium-sized air ways.
- The normal components of the wall, including cartilage, muscle, & elastic tissue, are destroyed &may be replaced by fibrous tissue.

Causes

- pulmonary infections
- Obstruction of the bronchus
- Aspiration of foreign bodies & vomitus
- Pressure from tumors, dilated blood vessels & enlarged lymph nodes
- ***Predisposing factors:*** Infection in early childhood (measles, influenza, TB, & immuno -deficiency disorders .

Pathophysiology

- The inflammatory process associated with pulmonary infections damages the bronchial wall, causing a loss of its supporting structure and resulting in thick sputum that ultimately obstructs the bronchi.
- The walls become permanently distended by sever coughing and distorted, impairing mucociliary clearance
- The retention of secretions and subsequent obstruction ultimately cause the alveoli distal to the obstruction to collapse (atelectasis).

Pathophysiology …

- Inflammatory scarring or fibrosis replaces functioning lung tissue.

- In time, the patient develops respiratory insufficiency with reduced vital capacity and decreased ventilation. There is impairment in the match of ventilation to perfusion (ventilation–perfusion imbalance) and Hypoxemia.

Clinical manifestations

- Chronic cough with Copious amount of purulent sputum (up to 200ml)
- Hemoptysis (50-70%)
- Clubbing of the fingers and cyanosis on P/E : crackles, rhonchi, & wheezes may be heard

Diagnosis

- The definitive diagnostic evaluation is prolonged hx of productive cough and –ve for PTB.
- CT scan

Medical management

Objective

- is to prevent and control infection
- promote bronchial drainage

- Antibiotic based on the results of culture & sensitivity

- Postural drainage
- Surgical intervention, although used infrequently

Pneumonia

- Pneumonia is an inflammatory process of the lung parenchyma that is commonly caused by infectious agent.
- It is the most common cause of death.

Causes

- It can be caused by wide variety of etiological agents (infectious or non infectious) including

❖ Bacteria

❖ viruses

❖ Fungi

✓ Chemical (irritant gases)

✓ Radiation (>6 weeks of treatment of lung or breast CA with radiation.)

Mode of transmission

- Pathogens can be introduced in to the lungs and cause pneumonia in 3 different routes i.e.

✓ Aspiration

✓ Inhalation

✓ Circulatory / hematogeneous spread.

- Most often organisms that cause pneumonia enter to the lungs after being inhaled into the air ways.
- Inhalation of chemical substances can also cause it.

- Sometimes normally harmless bacteria present in the mouth may be aspirated in to the lungs /usually if the gag reflex is suppressed/ and caused pneumonia i.e. aspiration pneumonia.
- Predisposing factors for aspiration pneumonia are: coma / unconsciousness and Seizure.
- Pneumonia can also be caused from infections that spread to the lungs through blood stream from other organs i.e. hematogeneous spread. e.g. septicemia , endocarditis

Classification of pneumonia

A) According to the causative agents / Etiologic classification

✓ **Infectious / bacterial pneumonia** - are the most common & most important

- Bacteria can be gram negative or gram positive bacteria

Gram positive	Gram negative
• Streptoccocus pneumoniae	- E. coli
	- Klebsiella
• Staphylococcus aureus	- pseudomonas

Viral pneumonia:

- ❖ Influenza virus type A (is the most common)
- ❖ cytomegalovirus (in immune compromised pt)

Fungal pneumonia:

- ✓ histoplasma capsulatum (histoplasmosis)
- ✓ pneumocystis carinii pneumonia (PCP) (common in immune compromised pt and the mortality rate is high)

Parasitic pneumonia

Ideopathic(non infectious)

B) By anatomical dissemination/ Radio logic /Anatomic/ Classification

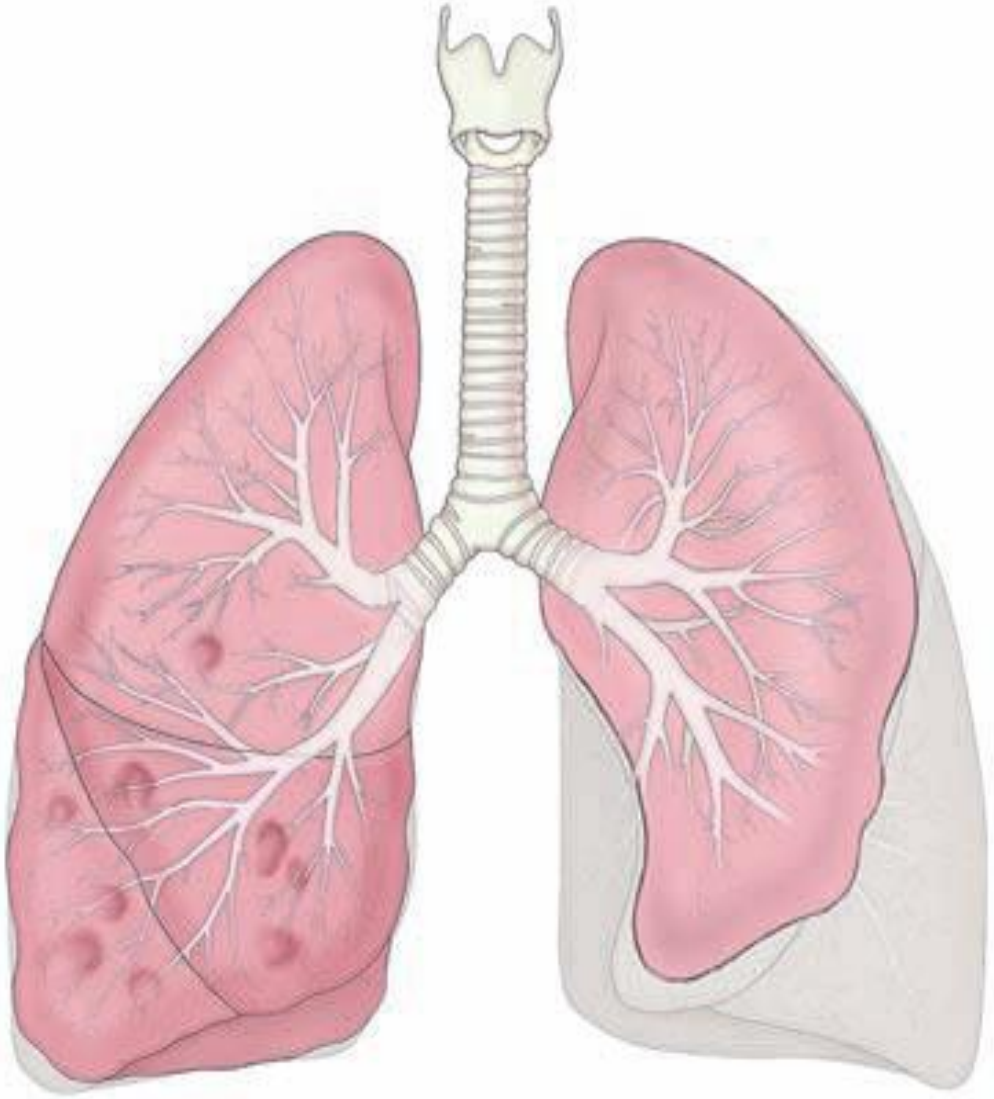
Lobar pneumonia

- ✓ If a substantial portion of one or more lobes is involved

Broncho Pneumonia

- ✓ It is distributed in a fashion, having originated in one or more localized areas with in the bronchi & extending to the adjacent surrounding lung parenchyma.
- ✓ It is more common than lobar pneumonia

Interstitial pneumonia



Bronchopneumonia **Lobar pneumonia**

C) By the area or setting the diseases acquired

Community Acquired pneumonia / CAP/

- Pneumonia that occurred in the community /outside the hospital or is diagnosed with in the first 48 hrs after admission to the hospital.

- It is a pneumonia in a community.

✓ *Causative agent*

- ❖ Streptococcus pneumonia / 50 – 60% /
- ❖ Staphylococcus aureus
- ❖ H. Influenza

- Atypical pneumonia / 20%/
  - ➢ Ricktesia
  - ➢ Chlamydia
  - ➢ Mycoplasmas
  - ➢ Fungus

Hospital Acquired pneumonia/nosocomial / pneumonia

- ✓ Is pneumonia w/c occurs 48 hrs or more after admission (mostly5-7 days of hospitalization)

Causative agents

- ❖Multi resistant bacteria
- ❖Staph. Auerus
- ❖Pseudomonas spp.
- ❖Gram negative enteric bacilli / e. Coli/

D) On the basis of clinical presentation / Clinical classification

✓ **Typical pneumonia**

- Pneumonia with sever manifestations
- Usually caused by bacteria

Clinical Manifestations

- Abrupt onset of fever
- Chills
- Productive Cough with purulent sputum
- Grunting , nasal flaring and use of accessory muscles.
- Sever/pleuritic type of chest pain
- On physical examination : dullness to percussion on the affected area, crackles and bronchial breath sounds
- Chest x-ray may show patchy infiltrate and small pleural effusion in some pts and
- Elevation of WBC

✓Atypical pneumonia

- Produce less striking symptoms and physical findings than bacterial pneumonia
- i.e. dry cough with mucoid sputum, generally no fever, few scattered wheezes and crackles, WBC commonly < 10,000u/l and chest x ray shows minimal or no infiltration.

Clinical Features of pneumonia

Cough : - first dry then rusty sputum / purulent/ after 24 hours

- Fever
- Pleurtic chest pain / aggravated by inspiration /
- Tachycardia
- Bronchial breath sound, dullness & some times crepitation

Diagnosis

- P/E: wheezs, crackles, dulness
- WBC & differential count
- Sputum gram stain
- Chest x – ray

Markers of severe pneumonia

- Respiration rate greater than 30/ min / adult/
- Cyanosis and Hypotension
- Confusion
- More than or equal to two lobe is involved

Complications

- Pulmonary effusion
- Empyema /Lung abscess
- Lobar collapse
- Sepsis

Managements

✓ **General Management: -**

- Analgesics for pain
- O_2 administration
- Antipyretics
- Fluid mgt

✓ Specific Management: -

- **Community acquired ambulatory pts /mild pneumonia**
  - Amoxicillin 500mg po. Tid 7 days OR
  - Erytromycine 500 mg PO QID for 5-7 days or
  - Ampicillin 500mg po. Qid for 7 days
- Alternative
  - Procaine penicillin 800,000IU IM daily for 5-7 days.

- ✓ **Community acquired hospitalized pts/ For severe CAP**

- Admit the patient
  - Bed rest
  - Oxygen administration
  - Analgesics for chest pain
  - Fluid and nutrition replacement
  - Treat fever

- Frequent monitoring of temperature, BP and PR.
- Crystalline penicillin 150,000 – 250.000 lu/ kg/ 24 hours, divided in to 6 doses.
- Or benzyl penicillin IV 2 million IU 6 hrly for 7-10 days **Plus** Gentamycin 5-7 mg/kg IV daily divided doses for 7 days
- Or Doxycyclin 200mg PO immediately followed by 100mg BID for 7-20 days. Or Erytromycin 500mg PO every 6hrs for 7-10 days

✓ **Hospital acquired pneumonia**

- Cloxacillin 1-2gm IV every 6hrs for 7 days plus Gentamycin (5-7mg/kg) 80mg iv / Im tid for 7 – 10 days or

- 3rd generation Cephalosporin e.g. Ceflazidime 1gm IV every 8 hrs

- Or Ceftriaxione 1-2gm IV/IM BID for 7days **plus** either Gentamycin or Ciprofloxacin 500mg PO /IV BID for 7 days.

✓ **Aspiration pneumonia**

- Clindamycin 600mg iv every 6 to 8 hours or
- Metronidazole 500mg po tid &
- Amoxicillin 500mg po / iv tid

✓ **Pneumonia in immuno - compromised**

- **pneumocystic carnii pneumonia / pcp/**

Clinical Manifestations

  - shortness of breath / SOB/
  - dry cough
  - Fever
- **P/E**: - Marked teachypnea & cyanosis
  - chest finding are minimal
- **RX**:- co – trimoxazole 480mg 4 tabs tid for 21 days

Nursing intervention

- ✓ Maintain a patent airway and adequate oxygenation.
- ✓ Obtain sputum specimens as needed.
- ✓ Use suction if the patient can't produce a specimen.
- ✓ perform chest physiotherapy.
- ✓ Provide a high calorie, high protein diet of soft foods.
- ✓ To prevent aspiration during nasogastric tube feedings, check the position of tube, and administer feedings slowly.

Nursing intervention …

- ✓ To control the spread of infection, dispose secretions properly.
- ✓ Provide a quiet, calm environment, with frequent rest periods.
- ✓ Monitor the patient's ABG levels, especially if he's hypoxic.
- ✓ Assess the patient's respiratory status. Auscultate breath sounds at least every 4 hours.
- ✓ Monitor fluid intake and output.
- ✓ Evaluate the effectiveness of administered medications.
- ✓ Explain all procedures to the patient and family.

Preventive measures

- Pneumococcal vaccine
- Frequent turning of bed ridden patients and early ambulation as much as possible.
- Coughing and breathing techniques.
- Sterilization of respiratory therapy equipment
- Suctioning of secretion in the unconscious who have poor cough and swallowing reflexes, to prevent aspiration of secretions and its accumulation.

Atelectasis

- Def^{n} It is collapse of an alveolus, a lobule, or larger lung units.

✓ **Causes**:

- Obstruction of a bronchus by wide variety of diseases e.g. pneumonia , asthma or foreign body, or a plug of thick exudates.
- Post operative complication is the most common cause, the onset is usually 24-72 hrs after operation,
- It may also resulted from pressure in the thorax caused by pleural effusion , pneumothorax etc.

Cont---

- ✓ **Risk factors:-**
- Respiratory depression from opioids, sedatives, & abdominal distention
- Post operative patients, bedridden patients
- Pleural effusion, pneumothorax

✓ **C/M**:

- Marked dyspnea
- Cyanosis
- Pleural pain
- Tachycardia and fever

Cont----

- ✓ **Medical Mgt:-**
  - Aspirate the plural effusion & pneumothorax
  - Force the pt to cough and hyperventilate,
  - Mechanical ventilation may be necessary
  - Remove the causes/removal of bronchial obstruction

- ✓ **Prevention:**
  - Encouraging the patient to cough, aspirate secretion, & postural drainage.
  - Turn the patient frequently.
  - Keep postoperative pts in their side.
  - Coughing & deep breathing exercise postoperative.

Pulmonary Emphysema

- ***Def^{n}***: - It is an abnormal distention of the air spaces beyond the terminal bronchioles with destruction of the wall of the alveoli.
- Is permanent distention and over aeration of the alveoli/air sacs
- The major changes that occur in the pulmonary emphysema is loss of lung elasticity and hyper inflation of the lung. i.e. airs are trapped in the lung.
- It is the end stage process that has progressed slowly for many years. It is irreversible problem

Cont---

✓ **Etiology**

- There is no clear causes but some factors may facilitating its development .

✓ **<u>Predisposing Causes</u>**: -

  - Cigarette smoking : - is the major cause
  - Air pollution
  - Infection

Cont---

✓ **C/M** :

- Dyspnea/main symptom
- Patient usually has a history of cigarette smoking and history of chronic cough
- Wheezing,
- Tachypnea
- The symptoms are exacerbated with a respiratory infection
- The onset is insidious
- hyper resonant on percussion,
- Anorexia, weight loss, & weakness

Cont---

✓ **DX**:

- History & physical examination
- Chest x – ray , blood gases.

Medical management

- ✓ **Bronchodilator**:- given to dilate air way & these medications include:
- Aminophylline, 5mg/kg by slow i.v push over 5 minutes
- Theophedrin 1 tab po tid
- ✓ **Aerosol:**
- Salbutamol aerosol inhalation 2 puff 3 – 4x/day

Cont---

✓ **Treatment of infection**

- Patients with emphysema are susceptible to lung infections& must be treated at the earliest signs of infection.
- The most common organisms are S.pneumonia & H. influenza.
- Ampicillin 500mg po Qid for 7 – 10 days OR
- Amoxicillin 500mg po Tid for 7 – 10 days OR
- CO – trimoxazole 960mg po bid for 7 days
- Oxygenation: Administer oxygen

Pleurisy/ pleuritis

- It refers an inflammation of both layers of the pleurae (parietal and visceral).

Clinical Manifestations

- Severe, sharp, '' knifelike'' pain during inspiration
  - friction rub
  - The pain may be localized or radiates to the shoulder / abdomen
- Pleurisy may develop with pneumonia, URTI, tuberculosis and pulmonary embolism

Diagnosis

- Chest X- rays
- Sputum examination
- Pleural fluid analysis
- Pleural biopsy

Medical Management

- Treat the underlying disease /pneumonia. . . /
- Analgesics / Indometacin 25mg p.o tid for 1week

Nursing Intervention

- Turn frequently on the affected side to splint the chest wall / this lessen the stretching of the pleura/.
- The nurse also can teach the patient to use the hands or a pillow to splint the rib cage while coughing.
- Emotional support & teaching

Pleural effusion

- Is an accumulation of fluid in the pleural space.

Causes

- Increase capillary permeability (e.g. infections pneumonia, TB , lung CA or trauma)
- Impair lymphatic function (e.g. lymphatic obstruction due to tumor)
- Increase systemic hydrostatic pressure (e.g. heart failure)

Clinical manifestations

- The size of effusion will determine the severity of the symptom.
- If the effusion is small its presence may be discovered only on chest x-rays and dyspnea may not be present.
- In large effusion lung expansion may be restricted
- Dyspnea with exertion
- Dry non-productive cough caused by bronchial irritation
- Decreased or absent tactile fremitus
- Percussion notes may be dull

Diagnosis

- Chest x ray
- Plural fluid analysis of :
    - WBC
    - RBC
    - Malignant cells
    - Microorganisms

Management

- Thoracenthesis
- Water seal chest drainage
- For fibro thorax (decortications involves removal or restrictive mass of fibrin & inflammatory cells)

Empyema

- It is a collection of purulent liquid / pus/ in the pleural cavity.

Causes

- It occurs as result of pneumonia or injury to the chest.
- It may also occur if the lung abscess extends through to pleural cavity.

C/M: -

- Fever, anorexia, & weight loss
- Night sweating
- Chest pain
- Dyspnea

Diagnosis

- History & physical examination
- On auscultation absence of breath sounds
- On percussion – dullness
- Chest x ray
- Pleural fluid analysis

Medical Mx: -

- Thoracentesis: - if fluid is not too thick
- chest drainage tube
- Antibiotic in large doses, are prescribed based on the causative organism

Nursing Intervention

- Resolution of empyema is prolonged process.
- Instruct breathing exercises
- Provide care specific to drainage of pleural fluid

Pulmonary Embolism

- Thrombosis : is a formation of solid mass with in the blood vascular system.
- The solid mass formed is called thrombus
- The thrombus may break off the wall of the vein and known as embolism, / is any mass lying free in the circulation.
- Pulmonary embolism refers to the obstruction of one or more pulmonary arteries by a thrombus.
- Is usually a post operative complication.

Risk Factors

- **Venous stasis** / slowing of blood flow in veins/ which may be due to:-
  - Prolonged immobilization / post operative/
  - Prolonged period of sitting
  - Varicose veins and Spinal cord injury

- **Hypercoagulability /** due to release of tissue thromboplastin often injury / surgery/ due to:-
  - -Injury
  - -Tumor
  - - Increased platelet count
- **Venous Endothelial Disease such as:-**
  - - Thrombophlebitis

- **Certain disease states such as:-**
  - Trauma
  - Postoperative / postpartum period
- **Other pre disposing factors include:-**
  - Pregnancy
  - Oral contraceptive use ???

C/M: - May be non specific

- Chest pain is the commonest symptom
- Sudden onset of dyspnea is the second most common symptom
- Tachypnea & Haemoptysis
- Tachycardia / rapid & weak pulse/
- Cough & diaphoresis
- Syncope & sudden death

- N.B. Multiple small emboli can lodge in the terminal pulmonary arterioles, producing multiple small infarctions of the lungs

Diagnostic evaluation

  - Chest x – ray
  - Pulmonary angiography

***Prevention*:-**

- Proper mgt of surgical pts /encourage early ambulation, deep breathing and coughing.
- Prevent deep vein thrombosis / active leg exercise to avoid venous stasis, early ambulation, & use of elastic stockings/.
- Anticoagulant therapy
- Avoid leaving IV catheter in veins for prolonged periods.
- Elevate legs above level of heart
- Advice against habits that increase venous stasis such as:-
  - Crossing leg
  - Sitting or lying down for prolonged period
  - Wearing constricting clothing

Medical Management

- The treatment of PE may include a variety of modalities:
  - General measures to improve respiratory and vascular status
  - Anticoagulation therapy
  - Thrombolytic therapy
  - Surgical intervention

Bronchial Asthma

- It is an intermittent, reversible, obstructive air way disease characterized by narrowing of the airways, resulting in **dyspnea, cough,** and **wheezing.**
- Asthma differs from other obstructive lung disease in that it is a reversible processes
- Acute exacerbations may occur, which last from minutes to hours, interspersed /followed by with symptom free periods.

- When asthma and bronchitis occur together, the obstruction is compounded and is called chronic asthmatic bronchitis. ‘’The pt fighting for air.’’
- If asthma becomes sever , prolonged /persistent and not respond to conventional therapy it is known as ***status asthmaticus.***
- Asthma can begin at any age.

Types of asthma

A) Allergic /extrinsic asthma

Is result from hyper sensitization of the bronchial mucosa by tissue specific antibodies.

- It is caused by known allergen/ dust, pollens, mold, etc. results in immediate antigen antibody reaction which release vaso-constrictive chemicals causing tissue damage,
- Most of the allergens are airborne & seasonal

- Usually seen in persons with family history of allergy and its onset occurs during child hood or adolescence.
- Persons with allergic asthma may also have past medical history of eczema or allergic rhinitis.
- Skin test for the offending allergens is positive.
- It is more common

B) Idiopathic or non allergic /intrinsic asthma

- It is not related to a specific allergens.
- Aggravating factors include common cold respiratory tract infections, exercise, emotions, & environmental pollutants.
- Some drugs, like aspirin, NSAIDs such as phenyl butazone, ibuprofen, and indomethacin may be a factor.

C) Mixed asthma

- It is the most common form of asthma
- It has the characteristics of both allergic & idiopathic asthma.

Pathophysiology

- The exact mechanism is not known but most of what is known involves the immunlnologic system and autonomic nervous system.
- Some pts with asthma develops exaggerated immune response to their environment.
- Persons with asthma react to low concentration of agents that do not normally cause broncho constriction. The obstruction is caused by one or more of the following:-

- Contraction of muscles surrounding the bronchi, which narrows the air way
- Swelling of membranes that line the bronchi
- Filling of the bronchi with thin mucus. In addition the bronchial muscles ad mucus glands enlarge; thick tenacious sputum is produced and the alveoli hyper inflate with air becoming trapped in the ling tissue.

- Initially allergens encounter
- Allergens stimulate plasma cells to produce antigen specific antibody/ IgE that binds to mast cells in air ways/ lungs.
- Re-exposure to the antigen results in the antigen binding to the antibody, /antigen antibody reaction causing the release of mast cell products called inflammatory mediators like - histamine, prostaglandin, and bradykinin
- Results in bronchospasm, membrane swelling and excessive mucus production

- In non allergic forms when nerve endings in the airways are stimulated by such factors as infection, cold air, exercise, smoking etc, large amount of acetyl choline (Ach) is released



- Ach results in broncho-constriction and production of mediators



- bronchospasm, membrane swelling and excessive mucus production

C/M

- The common symptoms of asthma are cough, Dyspnea and Wheezing
- Expiration is more strenuous and prolonged than inspiration?
- Asthma attacks often occur at night or early in the morning.
- Patients prefer sitting position during the attack
- The total attack may last 30 minutes to several hours and subside spontaneously.

Other symptoms include

- – Chest tightness
- – Prolonged expiration
- – Diaphoresis

✓ Late signs of poor oxygenation

- Tachycardia
- Widened pulse pressure
- Hypoxemia
- Cyanosis

DX:

- Complete history /Personal & or family history of allergic disease & P/E

- Chest x – ray :- over inflated lung

- Sputum & blood study (IGE ↑)
- N.B. pulmonary function usually normal b/n attacks but during an acute attack there is an increase total lung capacity and functional residual volume secondary to air trapping.

Treatment

- ✓ **Acute asthma attacks RX**
- Administer concentrated oxygen by mask (up to 6 liters/ min)
- Rehydrate the patient
- ✓ **Drugs Rx**
- Bronchodilators e.g. salbutamol,
- Anticholinergic, e.g. atropine and methyl nitrate
- Corticosteroids e.g. pridinsolone and mast cell inhibitors e.g. Cromolyn sodium inhalation.

✓ **Initial Rx:-**

A/ **First line**

- Salbutamol 2 puff/200mcgm & repeat after 20 minute for the first hour

OR

- Aminophylline, 5mg/kg slow iv push over 5 minutes. The same dose could be repeated after 30 minutes.

B/ Alternative

- Adrenaline, 1:1000, 0.5ml. Repeat after 30 minute to 1 hour if patient doesn't respond.

- ✓ If response to initial therapy is poor, give the following
- Aminophylline drip load dose 3-5 mg/kg in dextrose & water over 20 minutes. Then maintenance dose 0.6mg/kg hour in 5% D/W.

Plus

- Hydrocortisone, I.v 200mg stat and/or Prednisolone, 40 – 60mg po in divide dose immediately after hydrocortisone for 5 – 7 days.

- ✓ **RX of chronic asthma**
- **Intermittent Asthma**
- **First Line**: - salbutamol, inhalation 2 puff, 3 times a week
- **Alternative**: - ephedrine 1 tab 3 times per day + theophyline 100mg 2-3 times per day.

- ✓ **Persistent Asthma**
- ✓ **First Line**: - salbutamol inhalation 2 puff 3 times/ day

OR

- ephedrine 100mg TID **Plus** Beclomethasone 200mcg inhalation puff daily for two weeks and reduce to 100mcg if symptom improves.

OR

- Prednisolone, 0.5mg, po/ day.
- **Prevention:** avoid causative agent w/n ever possible.



Table 24-4 • Classifications of Medications Used in Asthma

MEDICATION CLASS	DRUG (BRAND NAME)
Long-Acting Control Medications	
Corticosteroids	beclomethasone (Beclovent, Vanceril), budesonide (Pulmicort), flunisolide (AeroBid), fluticasone propionate (Flovent), triamcinolone acetonide (Azmacort), prednisone
Mast cell stabilizers	cromolyn sodium (Intal), nedocromil sodium (Tilade)
Long-acting beta$_2$-adrenergic agents	salmeterol (Serevent), albuterol (sustained release) (Volmax ER), formoterol fumarate (Foradil)
Xanthine derivatives	aminophylline, theophylline (Slo-bid, Theo-24, Theo-Dur)
Leukotriene modifiers (inhibitors)	zafirlukast (Accolate), zileuton (Zyflo), montelukast (Singulair)
Combination products	albuterol sulfate + ipratropium bromide (Combivent); fluticasone propionate + salmeterol inhalation powder (Advair Diskus)
Quick-Relief Medications	
Short-acting beta$_2$-adrenergic agents	Albuterol (Proventil), levalbuterol (Xopenex), pirbuterol (Maxair), bitolterol (Tornalate)
Anticholingerics	ipratropium bromide (Atrovent)

Care of the patient with chronic obstructive pulmonary disease / COPD/

- **<u>COPD include</u>**: - chronic bronchitis, bronchiectasis, emphysema, & asthma.

✓ **Nursing diagnosis**

- Impaired gas exchange related to ventilation – perfusion inequality
- Ineffective airway clearance related to broncho constriction, increased mucus production

Nsg Dx cont---

- Ineffective breathing pattern related to shortness of breath, mucus
- Self – care deficits related to fatigue 2^0 to increased work breathing
- Activity intolerance due to fatigue, hypoxemia, &ineffective breathing patterns.
- Knowledge deficit about self – care procedures

Cont---

- **Goal (** Plan)
  - Improvement in gas exchange
  - Achievement of airway clearance
  - Improvement in breathing pattern
  - Independence in self – care activities
  - Improvement in activity tolerance
  - Compliance with therapeutic program & home care

Cont---

- **Nursing Intervention**
  - Administer bronchodilators as prescribed
  - Administer oxygen
  - Encourage fluid intake
  - Deep breathing exercise
  - Perform postural drainage
  - Encourage patient to begin to bathe self, dress self & walk
  - Support patient in establishing a regular regimen of exercise
  - Discuss with patient about drugs

Cont----

✓ **Potential Complication of COPD**

- Atelectasis
- Rib fracture
- Puemonia
- Pneumothorax
- Status asthmatics
- Respiratory failure

Pulmonary tuberculosis /PTB

- TB is an infectious diseases which primarily affects the lung (85%).
- It may also transmitted to other parts of the body i.e. pleura, spine, Meninges, kidneys, bones, lymph nodes etc.

✓ **Epidemiology:**

- It is estimated that one third of the world's population is infected with Mycobacterium TB
- Each year, about 9 million people develop TB, & 2 million die.
- About 1 million (11%) occur in children(< 15 years of age)

✓ **Etiology**

- Myco bacterium tuberculosis (Mb TB) which is aerobic bacteria, road shaped, acid fast bacillus.

✓ Other mycobacterium

- M. africanum(seen in western Africa)
- M. bovis(<1%, bovin TB)
- M.avium

Transmission

- Via air born (inhalation) , during coughing, sneezing, speaking produce tiny droplet→ dry out, suspended in the air for several hours infectious droplet are inhaled
- Consumption of raw milk containing M. bovis.

Factors contributing for better transmission

- closeness of contact
- poor ventilation
- duration of exposure,
- bacteriological status of pt

- ✓ **Risk factors**
- Person who has close contact with someone who has active TB
- Immune compromised persons e.g. HIV/AIDS, DM
- Poverty, mal-nutrition, pregnancy, alcohol, drugs, tobacco….
- Person living in overcrowded substandard housing
- Health care workers
- Age(child 5x higher risk than older)

Natural history

- In the great majority (90-95%) of persons infected with M.TB, but the immunological defense either kills or suppressed the inhaled bacteria causing latent M.TB infection.
- Only about 5-10% of such persons (primary infection) develop active TB.
- This reactivation leading to active TB w/c primarily affect the lung but can involve any other parts of the body.

- If untreated TB leads to deaths within 2-3 years in at least half of the patients.
- Without treatment about 20-25% would have natural healing and 25-30% would remain chronically ill, thus continuing to spread the diseases in the community.

✓ **IP**

- 4-16 weeks after infection

Over view of active and latent TB

	Active TB / TB diseases = a case of TB	Latent TB /TB infection not a case of TB , people w/o active TB
Definition:	TB bacteria is active in the body i.e. the immune system is unable to stop bacteria from growing	Bacteria are inside the body but not active. Body is able to fight off the bacteria and stop growing or eliminate but mostly the bacteria remain dormant & viable for years
Transmissions	Contagious	Not contagious
Symptoms	Develops symptoms	No symptoms
Chest X-ray	Usually abnormal	Normal
Sputum	Usually positive	Negative
TB skin test	Is usually positive	Is usually positive (b/c of TB bacilli in the body)

C/M of PTB

✓ General symptoms

- Fever
- Night sweating
- Chest pain
- weight loss
- Fatigue
- loss appetite/anorexia

✓ Pulmonary symptoms

- Cough (usually productive)
- Chest pain
- Dyspnea
- Hemoptysis

C/ms of Extra pulmonary TB (EPTB)

✓ C/M of EPTB depends of site of involvement e.g.

- **TB lymphadenitis**: slowly developing and painless enlargement of lymph nodes eventually drainage of pus.
- **TB meningitis:** headache, fever, vomiting, neck stiffness, mental status change.
- **Intestinal TB:** loss of apatite and weight , abdominal pain, diarrheal, fluid in the abdominal cavity (ascites)

Diagnosis

- Complete hx and P/E
- Sputum for AFB
- Culture (very sensitive and specific but expensive) and X-ray : for smear negative, children, & EPTB
- Tuberculin skin test
- Biopsy
- Blood (ESR, CBC.)

Tuberculin skin test

- Inject intradermally 0.1 ml of PPD tuberculin
- Read reaction 48-72 hours after injection
- Measure only indurations
- Record reaction in millimeters



Cont----





Interpretation

Tuberculin Skin Test (PPD)

≥ 5 mm induration	≥ 10 mm induration	≥ 15 mm induration
HIV (poor CMI) High suspicion (close contact, fibrotic CXR)	High prevalence ↓ reactivity (steroids, malnutrition)	All others (low probability)

Cont---

- ✓ **Chest x-ray:** is sensitive but less specific, because abnormalities can also be caused by other conditions
- ✓ **Suggestive findings**
- Upper lobe infiltration
- Cavitations (thick wall)
- Patchy, nodular shadows around the cavity

Diseases classification

A) Smear positive pulmonary TB

- Tb in a patient with at least two initial smear examination positive by direct microscope/DM for acid fast bacilli /AFB or
- Tb in a patient with one initial smear examination positive and positive by culture or
- Tb in a patient with one initial smear examination positive and x- ray abnormalities

B) Smear negative PTB

- Tb in patients with three initial smear examination negative by DM for AFB and who has failed to respond to a course of broad spectrum antibiotics or
- Three negative smear examination by DM and x-ray abnormalities or
- Three negative smear examination by DM but positive by culture

C) Extra pulmonary TB (EPTB)

- TB in organs other than the lungs, proven by one culture positive specimen from an extra pulmonary site or histo-pathological evidence from a biopsy.

Tb case definition by history of previous treatment

- **New patient:** have never had treatment for TB, or have taken anti-B drugs for less than 4 weeks. They may have disease at any anatomical site
- **Previously treated patient:** have received 4 weeks or more of anti-TB drugs in the past, may have disease at any anatomical site. They are further classified by the out come of their most recent course of treatment

Standard TB case definitions based on history of previous TB treatment

- **New case (N):** pt who never had treatment for TB or has been on previous anti TB Rx for less than 4 wks.
- **Relapse (R):** a pt declared cured or treatment completed of any form of TB in the past, but who reports back to the health service and is now found to be AFB smear-positive or culture positive.
- **Treatment after failure (F):** a patient who, while on treatment, is smear-positive at the end of the fifth month or later, after commencing

Cont---

- **Return after default (D):** a patient previously recorded as defaulted from treatment and returns to the health facility with smear-positive sputum
- It also includes a pt who was initially sputum smear negative but who becomes smear positive during the treatment.

Cont---

- **Transfer in (T):** a patient who is transferred in to continue treatment in a given treatment unit after starting treatment in another treatment unit. The patient must have taken anti-TB treatment for at least 4 weeks in the transferring health facility. The receiving treatment unit should register such patients as “transfer in” or “T” in the unit TB registers.
- **Other (O):** a patient who doesn’t fit in any of the above mentioned categories (e.g. a PTB smear negative who returns after treatment interruption

Treatment of TB

✓ Objectives

- To cure the pt from TB
- To prevent death from active TB
- To prevent relapse (by eliminating the dormant bacilli)
- To prevent the development of drug resistance (by using a combination of drugs)
- Decrease TB transmission to others

Drug Management

✓ First line essential anti TB drugs in Ethiopia

1. S= Streptomycin
2. R= Rifampicin
3. H= Isoniazid(INH)
4. Z= Pyrazinamide
5. E= Ethambutol

✓ **Drugs available in fixed dose combination (FDC)**

- E + H = EH (400/150mg)
- R + H = RH (150/75mg)
- R+H+Z = RHZ (150/75/400mg)
- R+H+Z+E = RHZE (150/75/400/275mg)

✓ **Drugs available as single drug:**

- E=400mg, INH= 100mg and 300mg S= 1gm

❖ Streptomycin is administered by injection while the other anti TB drugs are to be taken orally.

Phases of Rx

- **Intensive /initial phases:** the 1st 2 or 3 months. This phase consists of treatment with combination of four drugs for the first 8weeks for new cases, and with combination of five drugs for the first 8 weeks followed by four drugs for the next four weeks for re-treatment cases.
- **Continuation phase**: the remaining duration of Rx. This phase requires treatment with a combination of two drugs, to be taken for 4 months for new cases and treatment with a combination of three drugs for re-treatment cases for 5 months

Standard drug regimens

- ✓ There are two recommended standard of TB drug regimen:

1) Directly observed treatment short course (DOTS)
2) Long course chemotherapy (LCC)

Standard TB treatment regimen

- **New patient: 2(RHZE)/4(RH).** Other previously treated smear negative PTB and EPTB cases (case definition 'other') who were previously cured or treatment completed will be treated with new TB patient regimen
- **Previously treated:** previously treated cases will be re-treated with:**2S(RHZE)/1(RHZE)/5(RH)E.** Defaulted patient coming with smear negative TB, EPTB, or previously treated patient with unknown treatment outcome (case definition 'other')

Drug resistant TB

- **Mono resistant:** TB in patients were resistant to one first line anti TB drugs
- **Poly resistant:** TB resistant to more than one first line drug other than Isoniazide and Rifampicin.
- **Multi drug resistant TB (MRD-TB):** is active TB in w/c the organism is resistant to at least both INH and R the two most powerful anti TB drugs

Cont---

- **Extensively drug resistant tuberculosis (XDR-TB):** resistant to isoniazed and rifampicin (i.e. MDR) as well as any fluoroquinolone, and any of the second line injectable Anti-TB drugs (capreomycin, kanamycin and amikaccin)

- ✓ **Possible Cause of drug resistance**
  - Inadequate treatment of TB
  - Irregular treatment
  - Taking medication to short
  - Poor adherence to drugs etc

Multi drug resistant TB (MRD-TB)

- In Ethiopia it is estimated that MDR TB represents 1.6 % of new cases and 11.8% of retreatment cases.
- The diagnosis of MDR –TB is made only by reference laboratories performing culture for Tb strains with additional testing of anti TB drug sensitivity (drug sensitivity testing= DST)

Indication for DST

- Retreated cases who fail to respond to category II regimen
- Chronic cases
- New cases that are contact cases to known or suspected MDR cases.
- Unclear treatment failure with good compliance.

Treatment of MDR-TB

- Treatment is complex, very long and expensive, severe side effects are quite common and success rate is lower than with common TB (non MDR)
- MDR TB patients should be treated with a combination of second line drugs and first line TB drugs that still have efficacy
- The treatment should be standardized or individually adjusted on the basis of result of DST.

- The first phases which are injectable drug should be a minimum of 6 month and initial treatment may be extended if the patient does not convert both smear and culture.
- The entire treatment period is 18 – 24 months after smear and culture

Classification of anti TB drugs

Grouping	Drugs
G 1 = first line oral anti TB drugs	INH, Refampicin (R), Ethambutol (E), pyrazinamide (Z)
G2= injectable anti TB agents	Streptomycin (S), kanamycin (Km), Amikacin (Am), capreomycin (Cm), Viomycin(Vi)
G 3= Fluoroquinolones	Ciprofloxacin (Cfx), Ofloxacin (Ofx), Levofloxacin (lfx), Moxifloxacin (Mfx), Ganifloxacin (Gfx)
G 4= oral bacteriostatic second line agents	Ethionamide (Eto), Protionamide (Pto), Cycloserine (Cs), Terizidone (Trd), para aminoaslicylic acid (PAS), Thioacetazone (Th)
G 5 = anti TB agents with unclear efficacy	Clofazimine, amoxicillin, clarythromycin, linezolid. N.B. are not recommended by WHO

Tuberculosis and HIV

✓ **Challenges of HIV and tuberculosis co-infection**

- TB and HIV infections fuel each other.
- HIV infection does not only affect the prevalence and incidence of TB disease through reactivation and acquisition of infection but also modifies the clinical presentation and also complicates the treatment outcome of both diseases.

Impact of TB on HIV includes

- Is one of the leading and causes of illness and death among PLWHIV
- TB increases the occurrence of other Ois
- TB hasten the rate of HIV progression
- TB influence HIV on drug –drug interaction, side effects
- Late TB dx contributes increased death rates in PLWHIV

- So it is very important to diagnose and treat HIV-positive TB patients as soon as possible. i.e. every TB pt should tested for HIV and every HIV patients should screen for TB.
- When a person with HIV develops TB disease, we call this "**TB-HIV co-infection**".
- If these patients get HIV care and treatment too. This is referred to as "**TB-HIV co-management.**"

✓ TB-HIV Co-management at first level facility

- TB is an opportunistic infection that occurs frequently in HIV-positive patients.
- The general principle in the management of OIs in HIV-positive patients is that OIs should be treated before starting ART.
- However, TB is a special OI. The duration of treatment for TB is usually at least eight months, and during this time, dangerous OIs can occur if the patient has very low immunity.

- On the other hand, drug interactions between TB and ARV drugs can occur, and the number of pills the patient has to take if both TB and HIV are treated at the same time, is very high.
- The interactions between ARV and TB drugs occur mainly during the intensive phase of TB treatment, when the regimen contains rifampicin.
- The interactions can damage the liver or make some ARV drugs less effective. Additionally, overlapping side effects of ARV drugs and anti-TB drugs makes TB-HIV co-management complicated.

If a TB Patient is not on ART and CD4 count not available:

Patient clinical status	How to manage
Smear-positive pulmonary TB only (no other signs of WHO clinical stage 3 or 4), If patient is gaining weight on anti-TB treatment, and has no any other clinical conditions.	**Start** TB treatment. Start Tb treatment. Reassess and decide on ARV initiation after 2 months.
Smear-negative pulmonary TB only (no other signs of WHO clinical stage 3 or 4) If patient is gaining weight on anti-TB treatment and no new clinical conditions	**Start TB treatment.** **Continue anti-TB treatment:** Reassess the patient after intensive phase of TB treatment to determine whether to start ART during TB treatment or after completing it.
Any Pulmonary TB and patient has or develops signs of WHO clinical stage 4 or other stage 3 defining illnesses	**Start** TB treatment. Consult/refer to doctor for decision on TB-ART co-treatment plan as soon as possible. ART needs to be started immediately.
Extra-pulmonary TB	**Start** TB treatment.Consult/refer to a doctor for decision on TB-ART co-treatment plan as soon as possible. ART needs to be started immediately.

If patient not on ART and CD4 is available

CD4	How to manage
If CD4 < 200/mm3	Start TB treatment. Refer now to or consult a doctor for ART co-treatment plan. This needs to be started as soon as TB treatment is tolerated (between 2 weeks and 2 months).
If CD4 between 200-350/mm3	Start TB treatment. Refer to or consult a doctor for ART co-treatment after intensive phase (unless other WHO clinical stage 3 or 4 conditions are present, in that case refer the patient at once).
If CD4 > 350/mm3	Start TB treatment. Defer ART until TB treatment is completed unless the patient is in WHO clinical stage 4.

Guidelines for treatment of patients presenting with TB before initiation of ART

	Recommendation	Preferred ARV regimen
CD4 count <200/mm^3	▪ Start TB treatment. ▪ Start ART as soon as TB treatment is tolerated (usually between 2 -8 weeks of TB treatment)[1]	EFV containing regimen is preferred.[2] However, if drugs are unavailable or there are problems with EFV (adverse effects with intolerance and risk of pregnancy) use triple nucleoside regimen may be used with caution([3]). If patient develops ABC hypersensitivity continue NVP but monitor liver function every month.

Cont…

CD4 count 200-350/mm^3	• Start TB treatment. • Start ART after 8 weeks (after intensive phase) of TB treatment	• Start NVP containing regimen if the continuation phase does not include rifampicin. • If a non-pregnant woman has CD4 of >250 use EFV • In pregnant women with CD4 >250 use triple NRTI containing ABC • Start EFV containing regimen if the continuation phase includes rifampicin
CD4 >350/mm^3	• Start TB treatment • Defer ART	• Re-assess eligibility for ART at 24 weeks clinically and immunologically, in the course of TB-treatment, at completion of TB treatment, or as indicated.

Cont…

CD4 not available	• Start TB treatment. • Start ART after 2-8 weeks TB treatment if patient has severe disease and/or other clinical indicators of advanced immune deficiency • Start ART after completion of intensive phase when patient is not seriously ill or other signs of advanced immune deficiency are absent[5]	. • Start NVP containing regimen [4] if the continuation phase does not include rifampicin. • If a non-pregnant woman has CD4 of >250 use EFV • In pregnant women with CD4 >250 use triple NRTI containing ABC • Start EFV containing regimen if the continuation phase includes rifampicin

Cont…

[1] Timing of ART initiation should be up to clinical judgment based on other signs of immunodeficiency indicating progression of HIV disease (Refer to Table 1). For TB patients in WHO clinical Stage IV, ART should be started as soon as TB treatment is tolerated irrespective of CD4 count.

[2] EFV containing regimens include d4T/3TC/EFV or ZDV/3TC/EFV.

[3] NVP (200 mg daily for 2 weeks followed by 200 mg twice daily) may be used in place of EFV in absence of other options.

[4] NVP containing regimens include: d4T/3TC/NVP or ZDV/3TC/NVP.

[5] Start ART if non-TB Stage IV conditions are present

Prevention and control of TB

- Improve socio-economic status (living condition, nutrition…
- Vaccination
- Health education
- Avoid over crowding
- Isolation
- Early Dx and Rx
- Screening
- Boil milk before consumption

Lung Cancer / Bronchogenic carcinoma/

✓ **Risk factors**

- Tobacco smoke
- Second – Hand smoke
- Air pollution
- Occupational Exposure: - chronic exposure to industrial carcinogens, such as arsenic, asbestos, radiation, etc.
- Low vitamin A intake?
- Tuberculosis?

Cont---

- **C/M**:
  - Cough/ with or without sputum production/
  - Wheezing
  - Blood – tinged sputum
  - Chest pain / late manifestation/
  - Dyspnea
  - Symptoms of pleural & pericardial effusion

Cont----

✓ **Diagnostic Evaluation**

- Chest x – ray
- Cytologic examination of fresh sputum
- Lung scans

✓ **Medical Mx: -**

- Surgery: - Surgical resection
- Radiation Therapy
- Chemotherapy

Cont---

- **Nursing Intervention**
  - Remove Secretion from airway
  - Deep breathing & coughing exercise
  - Oxygen therapy
  - Relieve anxiety

Many thanks!!!